Math 152 Lecture Notes: Week 6

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0.1 When is a function invertible?

When we ask for an inverse for a function, it is important to specify where the function and its inverse will be defined. We write $f : X \to Y$ for a function taking input in X to outputs in Y. Then an inverse of f is a function $g : Y \to X$ such that $g(f(x)) = x$ and $f(g(y)) = y$ for $x \in X$ and $y \in Y$.

Last week, we saw that an inverse exists if and only if f is *injective* and *surjective*:

- A function is injective if it does not send two different inputs to the same output. Whenever $x_1 \neq x_2$, $f(x_1) \neq f(x_2)$. Equivalently, if $f(x_1) = f(x_2)$, then it must be that $x_1 = x_2$.
- A function $X \to Y$ is surjective if every element of Y is an output of the function. That is, for every $y \in Y$ there is an $x \in X$ such that $f(x) = y$.

Example 1. The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$ is both injective and surjective. It has inverse $g(y) = y^{1/3}$.

Example 2. The formula $f(x) = 1/x$ defines a function $\mathbb{R} \setminus \{0\} \to \mathbb{R}$. This function is injective but not surjective: 0 is not an output of the function. If we instead define f to be a function $\mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$, then it is invertible. In fact, it is its own inverse.

Example 3. The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - x + 1$, is surjective (we will show this below), but is not injective, as $f(0) = f(1) = 1$.

In general, given a function f, we can consider the equation $f(x) = y$. If for any y, this has a solution, f is surjective. If solutions are unique whenever they exist, f is injective. If both are true, and we solve for the unique solution x in terms of y, this gives a formula for the inverse function.

Example 4. Let's find an inverse for the function $f(x) = 2x^3 + 1$, which is injective and surjective as a function $\mathbb{R} \to \mathbb{R}$. We set up the equation $y = 2x^3 + 1$ and solve for x in terms of y:

$$\rightsquigarrow 2x^3 = y + 1 \rightsquigarrow x^3 = \frac{y+1}{2} \rightsquigarrow x = \left(\frac{y+1}{2}\right)^{1/3}$$

so our inverse is $g(y) = \left(\frac{y+1}{2}\right)^{1/3}$.

For many functions, it is not straightforward to determine the existence and uniqueness of solutions to these equations, and it may be intractable to explicitly solve for x in terms of y. At the end of last week, I stated criteria for injectivity and surjectivity that allow us to get around this using techniques of calculus.

Proposition. *Suppose that I is an interval and that $f : I \to \mathbb{R}$ is differentiable. If $f'(x) > 0$ for all x, then f is strictly increasing and so is injective. If $f'(x) < 0$ for all x, then f is strictly decreasing and is also injective.*

Proposition. *Suppose that I is an interval and that $f : I \to \mathbb{R}$ is continuous. Suppose that f is unbounded from below and from above, meaning that for any $a > 0$ there is x_1 such that $f(x_1) < -a$ and x_2 such that $f(x_2) > a$. Then f is surjective.*

Example 5. Let's show that the function $f : \mathbb{R} \to \mathbb{R}$ $f(x) = x^3 - x + 1$ considered above is surjective. It suffices to show that it is unbounded from below and from above. It helps to write

$$f(x) = x^3(1 - x^{-2} + x^{-3})$$

which holds when $x \neq 0$. If we choose $x > 2$, then $1 - x^{-2} + x^{-3} \geq 1/2$, and so $f(x) \geq x^3/2$. We can make this as large as we want by taking x sufficiently large, so f is unbounded from above. Similarly, if $x < -2$, then $1 - x^{-2} + x^{-3} \geq 1/2$, and $f(x) \leq x^3/2$ (the direction of the inequality is reversed because x^3 is now negative.) This can be made as small as we want, so f is unbounded from below as well. An argument like this can be used to show that any polynomial of odd degree is surjective.

Example 6. Consider the function $g : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = -x^5 - x$. This has derivative $g'(x) = -5x^4 - 1$, which is negative everywhere, so f is strictly decreasing and thus injective. This function is also surjective, for essentially the same reason as the previous one. Therefore, it is invertible. Its inverse is the Bring radical which I mentioned on Friday.

1 More on Inverses

1.1 The graph of the inverse function

The graph of a function f consists of all points $(x, f(x))$. If f has an inverse, then its graph consists of points $(x, f^{-1}(x))$. Writing $y = f^{-1}(x)$, we have $x = f(y)$, so the graph consists of points $(f(y), y)$. This is to say, it is obtained from the graph of f by switching the x and y coordinates. Geometrically, this corresponds to reflecting the graph over the line $y = x$ [Draw picture.]

Example 7. Graph $x^3 + 1$ and $(x - 1)^{1/3}$.

1.2 The derivative of the inverse function

If a function is continuous and differentiable, and has an inverse, then its inverse inherits these properties, at least partially.

Proposition. *Suppose that f is a continuous injective function from an interval I to an interval J. Then $f^{-1} : J \to I$ is continuous.*

If f^{-1} is also differentiable, we can find what its derivative must be. For notational simplicity, write $g = f^{-1}$, and consider the equation:

$$f(f^{-1}(x)) = x$$

Taking the derivative of both sides, using the chain rule, we obtain:

$$f'(g(x))g'(x) = 1$$

and therefore

$$g'(x) = \frac{1}{f'(g(x))}$$

at least if both sides make sense. It turns out that the left side makes sense wherever the right side does.

Proposition. *If f has inverse $g = f^{-1}$, and $f'(g(a))$ exists and is nonzero, then $g'(a)$ exists and is equal to $1/f(g(a))$.*

Remark. Because the derivative of a function is a "local" construction, depending only on the function near the point, g only needs to be an inverse of f on some interval containing a.

Example 8. Let's consider the case of an odd power $f(x) = x^n$. This is invertible as a function $\mathbb{R} \to \mathbb{R}$, with inverse $g(y) = y^{1/n}$. We already know what the derivatives of both of these functions are, but let's compute $g'(y)$ using the above formula to make sure that it works. We have $f'(x) = nx^n$, and so

$$\frac{1}{f'(g(y))} = \frac{1}{n(y^{1/n})^{n-1}} = \frac{1}{n} y^{-\frac{n-1}{n}} = \frac{1}{n} y^{\frac{1}{n}-1}$$

This agrees with the power rule for fractional powers. Notice that this expression is undefined for $y = 0$. Indeed, $y^{1/n}$ is not differentiable here, the graph has a vertical tangent line. [Draw picture.] This corresponds to a horizontal tangent line to the graph of x^n.

Later on, we will apply these considerations to inverse trigonometric functions. Before that, though, we will introduce the logarithm and exponential functions.

2 The Natural Logarithm Function

Our starting point with the logarithm will be to define it so that it is an antiderivative of $1/x$, and then study its remarkable properties.

2.1 The Logarithm as an Accumulation Function

Recall that we have an antiderivative of any power function x^q, where q is rational, except $q = -1$. The fundamental theorem of calculus gives us a way to construct and antiderivative of $1/x$: consider the accumulation function

$$\ln(x) = \int_1^x \frac{1}{t} \mathrm{d}t$$

[Draw picture] Because $f(t) = 1/t$ is continuous on $(0, \infty)$, this makes sense as a function $L : (0, \infty) \to \mathbb{R}$, and is continuous and differentiable with $\ln'(x) = 1/x$ there. (Later, we will consider antiderivatives of $1/x$ for x negative.)

Using this function, we can express lots of integrals that we could not previously express.

Example 9. Consider the integral

$$\int_0^1 \frac{x}{1+x^2} \mathrm{d}x$$

Using the substitution $u = 1 + x^2$, this is

$$\frac{1}{2} \int_1^2 u^{-1} \mathrm{d}u = \frac{\ln(2)}{2}$$

2.2 Algebraic Properties of the Logarithm

On Problem Set 3, you used the change of variable of variable formula to show that

$$\ln(ab) = \ln(a) + \ln(b)$$

for $a, b > 0$. We can repeatedly apply this to get a formula for L applied to products of more than 2 things:

$$\ln(a_1 \dots a_n) = \ln(a_1) + \cdots + \ln(a_n)$$

We will now study some consequences of this property for the function ln, which explain why we regard ln as a logarithm in the sense of solving $a^x = b$.

Proposition. *Let $a, b > 0$.*

1. $\ln(1) = 0$.
2. $\ln(1/a) = -\ln(a)$.
3. $\ln(a/b) = \ln(a) - \ln(b)$.
4. $\ln(a^n) = n \cdot \ln(a)$ *for any natural number* n.
5. $\ln(a^{p/q}) = \frac{p}{q} \cdot \ln(a)$ *for any rational* p/q.

Proof. For (1), $\ln(1)$ is an integral from 1 to 1, so is 0. For (2), we have

$$\ln(a) + \ln(1/a) = \ln(1) = 0$$

For (3), $\ln(a/b) = \ln(a \cdot (1/b)) = \ln(a) + \ln(1/b) = \ln(a) - \ln(b)$. For (4), write $a^n = a \cdot a \dots a$ with n a's, and get

$$\ln(a^n) = \ln(a) + \cdots + \ln(a) = n \ln(a)$$

For (5), we have $(a^{p/q})^q = a^p$, and so

$$q \cdot \ln(a^{p/q}) = p \cdot \ln(a)$$

from (4). Solving for $\ln(a^{p/q})$, we get (5). □

We can use these properties to simplify expressions for integrals using ln.

Example 10. Consider the integral

$$\int_2^5 \frac{1}{x} \mathrm{d}x$$

Because ln is an antiderivative of $1/x$, this is $\ln(5) - \ln(2) = \ln(5/2)$.

We can also use these properties to show that ln is invertible.

Theorem 1. *As a function* $\ln : (0, \infty) \to \mathbb{R}$, $\ln$ *is injective and surjective.*

Proof. To show that ln is injective, we notice that $\ln'(x) = 1/x$ is positive for any $x \in (0, \infty)$, so ln is strictly increasing.

To show that ln is surjective, it suffices to show that it is unbounded in both directions. First of all, consider

$$\ln(2) = \int_1^2 \frac{1}{x} \mathrm{d}x$$

This number is positive. We can see this in two ways. First, ln is strictly increasing, so $\ln(2) > \ln(1) = 0$. Second, we can actually estimate $\ln(2)$: for $x \in [1, 2]$, we have $1/2 \leq 1/x \leq 1$, and so $1/2 \leq \ln(2) \leq 1$ as the interval has length 1. [Draw picture] Now $\ln(2^n) = n \cdot \ln(2)$ for any integer n. [Draw picture] As $\ln(2)$ is positive, we can make this as large or as small as we want. Thus ln is unbounded from above and from below, and so is surjective because it is continuous. □

Definition. The number e is the unique positive number such that $\ln(e) = 1$. This makes sense because ln is injective and surjective, so this equation has a unique solution. It turns out that this number is irrational (in fact transcendental.) It is approximately equal to 2.7.

Later this week, we will discuss the inverse of the logarithm function, the natural exponential function, and say more about the number e. At this point, we have enough information to explain how to regard ln as a logarithm.

We can compute, for rational p/q:

$$\ln(e^{p/q}) = \frac{p}{q}\ln(e) = \frac{p}{q}$$

Thus, we can regard the function ln as a 'logarithm with base e.' We will see later that general exponential and logarithm functions with arbitrary real number exponents can be defined in terms of L and its inverse. I emphasize here that while rational powers can be defined in terms of repeated multiplication and extracting roots, we need to introduce these transcendental functions to define a^r for r an arbitrary real number.

2.3 Graphing the Logarithm Function

What does the graph of the logarithm function look like? Its derivative, and the proof that it is surjective, give us some clues. Because $\ln'(x) = 1/x$, which is positive for $x > 0$, very large for x near 0, and very small for x large, the function $\ln(x)$ is increasing everywhere, very rapidly near 0 and very slowly as x gets larger. Nevertheless, as $x \to \infty$, $\ln(x)$ gets arbitrarily large. We saw that powers of two give evenly spaced values of ln. [Draw picture, labeling points.]

Historical Digression

A variant of the natural logarithm was first introduced by John Napier in 1614. His motivation was that the property

$$L(xy) = L(x) + L(y)$$

allows one to reduce multiplication to addition in the following sense. Given a table of logarithms, if one wanted to multiply two numbers x and y, one could look up $L(x)$ and $L(y)$, add them, and then look up the number whose logarithm was this sum. Napier published a table of logarithms, and such tables were used as a computational tool for hundreds of years. The invention of the logarithm also led to the development of the slide rule, a calculating tool that is based on a logarithmic scale. These were widely used until the development of electronic calculators in the 1960s.

Napier's definition of his logarithm involved considering two particles moving along a line, such that one moves at a uniform velocity, while the other moves at a velocity proportional to its current position. In modern language, which we will learn later, this is describing a differential equation satisfied by the exponential function, the inverse of the logarithm. Our definition of the logarithm in terms of area under the graph of $1/x$ originates in 1649 with Alphonse Antonio de Sarasa and Grégoire de Saint-Vincent, who observed, as you did on Problem Set 3, that a function defined in terms of such areas turns multiplication into addition. The interpretation of logarithms in terms of finding a power, which is probably the definition you were taught previously, actually came much later.

2.4 Logarithmic Derivatives

As in its original historical context, the logarithm function can often be used to make it easier to deal with products, by turning them into sums. This is especially useful when differentiating large products.

The chain rule tells us that for f a function taking positive values:

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(f(x)) = \frac{f'(x)}{f(x)}$$

The right side of this makes sense as long as f is nonzero; we call it the logarithmic derivative of f, $\mathrm{D}\log(f) = f'/f$. Using properties of the logarithm or alternatively the product rule, we have

$$\mathrm{D}\log(fg) = \mathrm{D}\log(f) + \mathrm{D}\log(g)$$

More generally, wherever $f_1(x), \ldots, f_n(x)$ are all nonzero,

$$\mathrm{D}\log(f_1 \ldots f_n) = \mathrm{D}\log(f_1) + \cdots + \mathrm{D}\log(f_n)$$

and also

$$\mathrm{D}\log(f/g) = \mathrm{D}\log(f) - \mathrm{D}\log(g)$$
$$\mathrm{D}\log(f^n) = n \cdot \mathrm{D}\log(f)$$

That is, the logarithmic derivative operation turns multiplication into addition, as does the logarithm.

We can use these properties to compute derivatives of products, as follows:

Example 11. Consider the product $f(x) = x(x-1)(x-2)(x-3)$. For x not equal to $0, 1, 2, 3$, its logarithmic derivative is

$$\frac{f'(x)}{f(x)} = \frac{1}{x} + \frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3}$$

and therefore

$$f'(x) = \frac{f(x)}{x} + \frac{f(x)}{x-1} + \frac{f(x)}{x-2} + \frac{f(x)}{x-3} = (x-1)(x-2)(x-3) + x(x-2)(x-3) + x(x-1)(x-3) + x(x-1)(x-2)$$

2.5 More Integral Formulas

To conclude our discussion of the logarithm function, we will consider integrals expressible using the logarithm in a more systematic way. First of all, we can find an antiderivative for $1/x$ defined on the negative numbers as well.

Proposition. *The derivative of $f(x) = \ln|x|$ is equal to $1/x$ for any nonzero x.*

Proof. We already know that this is true for $x > 0$, where $|x| = x$. Suppose that $a < 0$. Near a, we have $f(x) = \ln(-x)$. Applying the chain rule, we have

$$f'(a) = -\frac{1}{-x} = \frac{1}{x}$$

□

This means that $\ln|x|$ is a an antiderivative of $1/x$ where it is defined, on $\mathbb{R} \setminus \{0\}$, or in other words:

$$\int \frac{1}{x}\mathrm{d}x = \ln|x| + C$$

This lets us apply substitution to express a large family of integrals in terms of the logarithm.

Proposition. *If f is a continuous function which is nonzero on $[a, b]$, then*

$$\int_a^b \frac{f'(x)}{f(x)}\mathrm{d}x = \ln|f(b)| - \ln|f(a)|$$

Proof. One can check that $\ln|f(x)|$ is an antiderivative, then apply the fundamental theorem of calculus. □

This can also be understood and applied in terms of u-substitution. Care should be taken, however, that the new introduced variable is actually nonzero between the bounds of integration

Example 12. Consider the integral

$$\int_2^3 \frac{x}{1-x^2}\mathrm{d}x$$

Taking $u = 1 - x^2$, $\mathrm{d}u = -2x\mathrm{d}x$, and this becomes

$$-\frac{1}{2}\int_{-3}^{-8}\frac{1}{u}\mathrm{d}u = -\frac{1}{2}\left(\ln|-8| - \ln|-3|\right) = -\frac{1}{2}\ln(8/3) = \ln(3^{1/2}/8^{1/2})$$

A particular application of this is to find an antiderivative of the tangent function.

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

As $\cos'(x) = -\sin(x)$, $\tan(x)$ has the form $-f'(x)/f(x)$, and we obtain

$$\int \tan(x)\mathrm{d}x = -\ln|\cos(x)| + C$$

Using the secant function $\sec(x) = \frac{1}{\cos(x)}$, this can be written $\ln|\sec(x)| + C$. This antiderivative can be used on intervals where $\tan(x)$ is consistent, that is to say, between zeroes of the cosine function. We similarly have

$$\int \cot(x)\mathrm{d}x = \ln|\sin(x)| + C$$

taking $u = \sin(x)$. This antiderivative can be used on intervals where $\sin(x)$ is nonzero.

Example 13. Consider the integral

$$\int_0^1 \frac{x}{\tan(x^2+1)}\mathrm{d}x$$

Taking $u = x^2 + 1$, this becomes

$$\int_1^2 \frac{1}{2\tan(u)}\mathrm{d}u = \int_1^2 \frac{\cot(u)}{2}\mathrm{d}u$$

The sine function is nonzero between 1 and 2, so this

$$\frac{1}{2}\left[\ln|\sin(u)|\right]_1^2 = \frac{\ln(\sin(2)) - \ln(\sin(1))}{2}$$

Later, when we study integration by parts, we will consider the indefinite integral of the logarithm function itself.

3 The Exponential Function

We have shown that the natural logarithm function is injective and surjective as a function $\ln : (0,\infty) \to \mathbb{R}$. This means that it has an inverse. We call this the natural exponential function, written $\exp : \mathbb{R} \to (0,\infty)$. Using our knowledge of the natural logarithm, we can say a lot about this function.

First of all, the graph of the exponential is obtained by reflecting the graph of the logarithm across the line $y = x$. [Draw picture].

As the graph depicts, as $x \to \infty$, $\exp(x) \to \infty$, and as $x \to -\infty$, $\exp(x) \to 0$. We can also obtain some values of the exponential function: $\exp(0) = 1$, and $\exp(1) = e$. [Label points] These follow from $\ln(1) = 0$, $\ln(e) = 1$.

3.1 Algebraic Properties of the Exponential Function

We can translate the algebraic properties of the exponential into algebraic properties of the logarithm.

Proposition. *Let a, b be real numbers, n an integer, p/q a rational number.*

1. $\exp(a+b) = \exp(a)\exp(b)$
2. $\exp(-a) = 1/\exp(a)$
3. $\exp(a-b) = \exp(a)/\exp(b)$
4. $\exp(na) = \exp(a)^n$
5. $\exp((p/q)\cdot a) = \exp(a)^{p/q}$

Proof Sketch. The proofs of all have the same form; I will illustrate (1). Using the algebraic properties of the logarithm, we have

$$\ln(\exp(a)\exp(b)) = \ln(\exp(a)) + \ln(\exp(b)) = a + b = \ln(\exp(a+b))$$

That is to say, the desired equation in (1) becomes true after we apply ln to both sides. Because ln is an injective function, this means that the equation in (1) must be true. □

In particular, this allows us to interpret values of exp as powers of e. Taking $a = 1$ in (5), we have:

Proposition. *For p/q rational, we have $\exp(p/q) = e^{p/q}$.*

Here $e^{p/q}$ has the usual meaning for a rational power, it is the qth root of e^p, where e^p is defined in terms of repeated multiplication. Soon, we will define exponents a^r where a and r are arbitrary real numbers using the exponential function, at which point it will make sense to say that $\exp(x) = e^x$. Notice that the properties of the exponential function observed above generalize identities involving integer or rational powers under this interpretation.

3.2 Derivatives and Integrals Involving the Exponential

Using the general formula for the derivative of an inverse function, we can find the derivative of the natural exponential.

Theorem 2. *The exponential function satisfies $\exp'(x) = \exp(x)$. This is to say, it is its own derivative.*

Proof. The general formula for the derivative of an inverse function tells us that so long as $\ln'(\exp(x))$ is well defined and nonzero, we have

$$\exp'(x) = \frac{1}{\ln'(\exp(x))}$$

Using $\ln'(y) = \frac{1}{y}$ and the fact that $\exp(x)$ is positive everywhere, we have

$$\exp'(x) = \frac{1}{1/\exp(x)} = \exp(x)$$

□

This fact accounts for much of the importance of the exponential function as we will see.

Example 14.

$$\frac{\mathrm{d}}{\mathrm{d}x}\exp(x^2) = 2x\exp(x^2)$$

by the chain rule.

Also using the chain rule, we obtain the generalization

$$\frac{\mathrm{d}}{\mathrm{d}x}\exp(kx) = k\exp(kx)$$

We will see the usefulness of this soon. As the exponential function is its own derivative, it is also an antiderivative of itself:

$$\int \exp(x)\mathrm{d}x = \exp(x) + C$$

This allows us to evaluate many integrals involving the exponential function.

Example 15. Consider the indefinite integral

$$\int \exp(2x-1)\mathrm{d}x$$

Using $u = 2x - 1$, $\mathrm{d}u = 2\mathrm{d}x$, this becomes

$$\frac{1}{2}\int \exp(u)\mathrm{d}u = \frac{\exp(u)}{2} + C = \frac{\exp(2x+1)}{2} + C$$

Example 16. Consider the indefinite integral

$$\int \frac{\exp(\sqrt{x})}{\sqrt{x}}\mathrm{d}x$$

Using $u = \sqrt{x}$, this is

$$2\int \exp(u)\mathrm{d}u = 2\exp(u) + C = 2\exp(\sqrt{x}) + C$$

Example 17. Consider

$$\int \exp(x)\sin(\exp(x))\mathrm{d}x$$

Using $u = \exp(x)$, this is

$$\int \sin(u)\mathrm{d}u = -\cos(u) + C = -\cos(\exp(x)) + C$$

4 General Exponential and Logarithm Functions

4.1 General Exponential Functions

We have seen that the exponential function, although ultimately defined in terms of an integral, generalizes rational powers of the number e: $\exp(p/q) = e^{p/q}$. Through a modification of this function, we can define a^r for any positive a and arbitrary real power r.

First, for p/q rational and $a > 0$, we can make the following calculation, using properties of the exponential and logarithm functions:

$$\exp((p/q)\cdot\ln(a)) = \exp(\ln(a^{p/q})) = a^{p/q}$$

This inspires us to make the following definition.

Definition. For $a > 0$, and $r \in \mathbb{R}$ an arbitrary real number, we define $a^r = \exp(r\cdot\ln(a))$.

Remark. Not only does this definition generalize the already established meaning of rational powers, it does so in such a way that the resulting function a^x is continuous and differentiable. This is why this generalization is distinguished.

Using the algebraic properties of the natural exponent and logarithm, we can generalize familiar algebraic properties of exponents:

Proposition. *Let $a > 0$, r, s arbitrary real numbers.*

1. $a^{r+s} = a^r \cdot a^s$
2. $a^{-r} = 1/a^r$
3. $a^{r-s} = a^r/a^s$
4. $(a^r)^s = a^{rs}$

Using this definition, we can define two new types of function. First of all, fixing r and replacing a with a variable x, we have a power function:

$$f(x) = x^r = \exp(r \ln(x))$$

This is a function $f : (0, \infty) \to (0, \infty)$.

Remark. The function $f(x) = x^r$ is algebraic if $r = p/q$ is rational, it satisfies $f(x)^q - x^p = 0$. If r is irrational, this function is transcendental.

Second, fixing $a > 0$ and replacing r with a variable x, we have an exponential function with base a:

$$g(x) = a^x = \exp(x \ln(a))$$

This defines a function $g : \mathbb{R} \to (0, \infty)$. Using the chain rule, we can find derivatives (and antiderivatives) of these functions.

Proposition. *The derivative of a power function x^r is*

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[x^r\right] = rx^{r-1}$$

This is to say, the familiar power rule applies to arbitrary real powers.

Proof. Writing $f(x) = x^r = \exp(r \ln(x))$ and applying the chain rule, we have

$$f'(x) = \exp(r \ln(x)) \cdot \frac{r}{x} = \frac{rx^r}{x} = rx^{r-1}$$

□

As a consequence, we have a general antiderivative of any power function

Proposition.

$$\int x^r \mathrm{d}x = \begin{cases} \frac{x^{r+1}}{r+1} + C & r \neq -1 \\ \ln(x) + C & r = -1 \end{cases}$$

Example 18.

$$\int_1^2 5x^{\sqrt{2}} = \left[\frac{5x^{\sqrt{2}+1}}{\sqrt{2}+1}\right]\Bigg|_1^2 = 5\frac{2^{\sqrt{2}+1} - 1}{\sqrt{2}+1}$$

Proposition. *The derivative of a general exponential function a^x is*

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[a^x\right] = \ln(a) \cdot a^x$$

Proof. Write $g(x) = a^x = \exp(x \ln(a))$ and apply the chain rule to obtain $g'(x) = \ln(a) \exp(x \ln(a)) = \ln(a) \cdot a^x$. □

This also gives us an antiderivative:

$$\int a^x \mathrm{d}x = \frac{a^x}{\ln(a)}$$

Example 19.

We can also differentiate powers where both the base and the exponent depend on x.

Example 20. Consider $f(x) = x^x$. (This is defined on $(0, \infty)$). To differentiate f, we write it as $\exp(x \ln(x))$ and apply the product rule and the chain rule:

$$f'(x) = \exp(x \ln(x)) \cdot \left(\ln(x) + \frac{x}{x}\right) = (\ln(x) + 1)x^x$$

I will leave it to you to find a general formula for the derivative of $f(x)^{g(x)}$.

4.2 General Logarithm Functions

For $a, b > 0$, $a \neq 1$ we define $\log_a(b)$ as

$$\log_a(b) = \frac{\ln(b)}{\ln(a)}$$

This satisfies

$$\log_a(a^x) = \frac{\ln(\exp(\ln(a)x))}{\ln(a)} = \frac{\ln(a)x}{\ln(a)} = x$$

and also

$$a^{\log_a(x)} = \exp\left(\ln(a)\right)$$

That is to say, $\log_a(x)$ is the inverse to a^x.

Using the formula, we can differentiate $\log_a(x)$:

Proposition.

$$\frac{\mathrm{d}}{\mathrm{d}x} \log_a(x) = \frac{1}{x \ln(a)}$$

Because $\log_a(x)$ is a constant multiple of $\ln(x)$, we will get antiderivatives of general logarithm functions when we get an antiderivative for ln.

Example 21. What is

$$\frac{\mathrm{d}}{\mathrm{d}x} \log_x(2)?$$

Writing

$$\log_x(2) = \frac{\ln(2)}{\ln(x)}$$

we have

$$\frac{\mathrm{d}}{\mathrm{d}x} \log_x(2) = \frac{-\ln(2)}{\ln(x)^2} \cdot \frac{1}{x} = -\frac{\ln(2)}{x \ln(x)^2}$$